STAC WARS

PS 2

SOLAR RADIATION

ANALYSIS AND PREDICTION

Voyage-123 decoded: read the sun, power the Rebellion



TRANSMISSION RECEIVED

A long time ago, in a galaxy far, far away…

The Rebellion's new hidden base runs on sunlight. Its panels hum all day, its comms crackle every time the sun flares, and nobody knows when either will happen.

Out past the Outer Rim, an old Alliance probe, Voyage-123, has been quietly logging the sunshine on a remote world: radiation, temperature, humidity, wind, sunrise and sunset. Luke Skywalker wants to know what the sun is really doing. R2-D2 has the logs, but they are a mess of raw numbers, and Artoo can only beep so many charts per day.

The catch: the next flare season is coming, and the base cannot afford a blackout. So Luke has called in the Alliance's sharpest data analysts. That's **you**.

> *"The sun gives us power, and sometimes trouble. If we can read it, we can plan for it." — Luke Skywalker*

YOUR MISSION

TOTAL: 15 MARKS

Voyage-123 has recorded data linked to solar flares in the galaxy. You must perform the following tasks:

Dataset: **Voyage-123 solar observation data (Google Drive)**

A. DATA VISUALIZATION

4 MARKS

> *"Graphs first, theories later. R2 insists."*

Using the given solar observation dataset, create appropriate visualisations to explore the data.

- Visualise the variation of solar radiation with time.
- Plot relationships between radiation and at least two other environmental parameters.
- Use appropriate plots to identify trends, patterns, or outliers.

> Bonus: Create an additional meaningful visualisation that reveals an interesting pattern in the dataset.

B. DRAWING INFERENCES

5 MARKS

> *"Even the sun has moods. Find out what they depend on."*

Based on the visualisations created in Part A, analyse the data and answer questions such as:

- During what time of day is solar radiation generally highest?
- Which environmental parameter appears to have the strongest relationship with radiation?
- What patterns can be observed between radiation, temperature and humidity?
- Are there any unusual observations or outliers? What might explain them?

Support your conclusions using evidence from the dataset.

> Bonus: Identify an additional non-obvious insight from the visualisations and justify it using the data.

C. SOLAR DAY ANALYSIS

5 MARKS

> *"Daybreak to dusk, one tidy report per day. No messy logs, says Artoo."*

Use Pandas and NumPy to transform the raw observations into a day-wise solar activity report.

Participants must:

- Convert the Data and Time columns into a proper datetime column.
- Group observations by date and calculate the daily maximum and average radiation.
- Determine the date with the highest recorded solar radiation.
- For that day, determine the time at which maximum radiation was recorded and report the corresponding temperature, humidity and wind speed.
- Compare the day's maximum radiation with its sunrise and sunset times and determine how long after sunrise the peak radiation occurred.
- Identify the top 5 days with the highest average radiation.

BONUS — SOLAR RADIATION PREDICTION

6 MARKS

> *"Teach a machine to see the sunrise before it happens."*

Build a machine-learning regression model to predict solar radiation using the available environmental and temporal features.

- Select appropriate input features.
- Split the data into training and testing sets.
- Train an appropriate regression model.
- Evaluate the model using RMSE and R^2 scores.
- Explain which features appear useful for predicting solar radiation.

SUBMISSION REQUIREMENTS

Submit your analysis containing:

1. Your .py/.ipynb files in the GitHub repository.
2. A report which must contain your plot, inferences and scores.

MAY THE CODE BE WITH YOU